



TÜRKIYE → GERMANY · PER DIN STANDARDS

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Before You Sell Welded Fabrication to Germany

10 Critical Questions to Ask Yourself

A field guide for Turkish metal and welding manufacturers

Why this guide?

Germany is one of the most attractive markets in the world for Turkish metal and welding manufacturers — and also one of the most demanding. Most Turkish producers make technically sound parts. The problem rarely lies in production; it usually appears in the bridge of **certificates, tolerance interpretation and communication**.

This guide distills the most expensive mistakes a Turkish-German team working in Germany has seen in the field into 10 questions. Ask yourself each one honestly. If you cannot answer even one of them clearly, you should close that gap before you ship to Germany.

Our goal is not to scare you — it is to help you **see expensive surprises before they happen**.

01

Which certificate is truly mandatory for my product — or just "nice to have"?

The most common and most expensive mistake: spending money and time on an unnecessary certificate, or skipping a mandatory one.

If your product is built into a German building, steel structure or load-bearing assembly, **EN 1090 (CE marking) is legally required**. For pure machined parts, the **ISO 3834** welding quality management system is often enough.

Check question: Which standard does your customer require in the contract? Don't start production before you have it in writing.

02

Do I know the correct execution class (EXC) for EN 1090?

EN 1090 has four execution classes, EXC1 to EXC4. The wrong choice means either unnecessary cost or a rejected batch.

EXC2 is typical for most standard steel structures. EXC3/EXC4 are required for critical structures such as bridges, high loads or dynamic stress. With each class, the requirements for inspection, traceability and welder qualification multiply.

Check question: Which EXC class does the customer require, and does your current workshop meet that class's traceability requirements?

03**Are my WPS and PQR documents truly valid and current?**

WPS (welding procedure specification) and PQR (procedure qualification record) are the first documents a German inspection looks at. This is where we see the most gaps in the field.

If your WPS is not based on a PQR qualified per EN ISO 15614-1, the document is invalid — no matter how good your welder is. Each WPS also has limits on material group, thickness range and position.

Check question: Do you have a valid, signed and dated WPS for every welded joint you produce?

04**Do my welders' certificates cover the work I actually produce?**

A welder qualification (EN ISO 9606) is not unlimited. It applies to a specific process, material group, thickness, diameter and position — and **it expires**.

Common problem: the welder has a TIG certificate but produces with MAG; or has a pipe certificate but welds plate. At inspection, the whole batch is rejected.

Check question: Does each welder's certificate cover the actual work (process + material + position + thickness), and is it still valid?

05**Can I read the weld symbols in the drawing correctly?**

German drawings use weld symbols per DIN EN ISO 22553. Arrow side or other side, a-dimension (a3, a5), all-around weld mark, grinding/machining note — every detail is binding for production.

Misreading a single symbol means the whole batch is produced with the wrong weld size. You often notice only once the part is already in Germany.

Check question: Can you interpret every weld symbol in the drawing without doubt? When unsure, asking the customer is cheaper than producing wrong.

06

Do I know the required weld seam quality level (ISO 5817)?

ISO 5817 defines three quality levels for weld seams: B (high), C (medium), D (basic). Depending on what the customer requires, the acceptance criteria change completely.

For level B, the permitted pores, undercuts and weld reinforcement values are much stricter. A seam produced for C will be rejected by a customer requiring B — even if it looks identical.

Check question: Which ISO 5817 level does the drawing or contract require, and does your quality control inspect to that level?

07

Are my material certificates (3.1 / 3.2) traceable?

The German customer usually requires a **3.1 inspection certificate** per EN 10204 (confirmed by the manufacturer); for critical work, **3.2** (confirmed by an independent party). This is not just paper — it is traceability down to the heat number.

Field mistake: the right material is used, but the certificate-to-part matching (heat number tracking) is missing. If at inspection you cannot say which certificate this part's material came from, the batch stops.

Check question: Do you match every material batch you use to its 3.1/3.2 certificate and keep a record?

08

Do I know the difference between S235JR and S235J2+N per product?

A very common and dangerous assumption: "S235 is S235." No. The suffixes **JR**, **J0**, **J2** indicate the temperature of the notch impact toughness; **+N** stands for normalizing.

If you send S235JR when the customer required S235J2+N, the material risks brittle fracture at low temperatures and the batch is rejected — and this is a serious safety error too.

Check question: Do you procure the material grade specified in the drawing down to its last letter and suffix?

09

Can I write an 8D report when a complaint arrives?

A complaint will arrive one day — that is normal. What matters is how you respond. The German customer usually requires an **8D report**: a structured problem-solving report with 8 disciplines.

A supplier who doesn't know 8D, or fills it in late/incompletely, gets stamped "unprofessional" — even when technically in the right. The immediate action (D3) in the first 24 hours and the root-cause analysis (D4, usually 5-Why or Ishikawa) set the tone.

Check question: If a complaint arrived tomorrow, could you respond within 48 hours with a clean 8D report?

10

Can I get the tone and language right with the German customer?

Being technically right is not enough; **how you communicate** matters at least as much as the product. German business culture expects clear, direct but respectful, timely and written communication.

Late replies, vague "we'll handle it" phrasing, or overly formal/wrongly-toned German e-mails erode trust. Most suppliers are lost not because of quality, but because of **communication trust**.

Check question: Are your e-mails to the German customer timely, clear and in the right German tone? If not, you need someone to build that bridge.

So what should you do now?

If you said "not sure" to even one of these 10 questions, you are not alone — most Turkish manufacturers selling to Germany are at the same point. The good news: these are all **learnable and closable** gaps.

As DINConnect, we are a Turkish-German team working in Germany. We don't issue certificates and we don't manufacture — we only build this bridge: the right certification roadmap, drawing interpretation and communication with the German customer.

If you would like to discuss any of the questions in this guide, write to us for a free initial consultation.

Free initial consultation

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